Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Vectors

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$. Find $\mathbf{a} \cdot \mathbf{b}$ and the angle between them. **[5]**

2. $\mathbf{a} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$. Find $\mathbf{a} \cdot \mathbf{b}$ and the angle. **[5]**

3. $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. Find $\mathbf{a} \cdot \mathbf{b}$. **[3]**

4. Find a vector equation of the line through $P(1, 0, -2)$ and $Q(3, -1, 4)$. Does $R(5, -2, 10)$ lie on it? **[5]**

5. Find a vector equation of the line through $A(2, 1, 0)$ and $B(4, -1, 3)$. **[4]**

6. Find a unit vector in the direction of $\mathbf{a} + \mathbf{b}$ where $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$. **[4]**

7. Find a unit vector in the direction of $3\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$. **[3]**

8. Points A, B, C have position vectors $\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $3\mathbf{i} - \mathbf{j} + \mathbf{k}$, $2\mathbf{i} + 4\mathbf{j} - \mathbf{k}$. Find angle BAC. **[6]**

9. Points A, B, C: $\mathbf{i} + \mathbf{j}$, $2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$. Find angle ABC. **[6]**

10. Show that the lines $\mathbf{r} = (\mathbf{i} + 2\mathbf{j}) + \lambda(\mathbf{i} - \mathbf{j} + \mathbf{k})$ and $\mathbf{r} = (3\mathbf{i} + \mathbf{j} + \mathbf{k}) + \mu(2\mathbf{i} + \mathbf{j} - 2\mathbf{k})$ are perpendicular. **[4]**

11. Find the point of intersection of $\mathbf{r} = (2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) + \lambda(\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ and $\mathbf{r} = (4\mathbf{i} + 5\mathbf{j} - \mathbf{k}) + \mu(2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$. **[6]**

12. Find the magnitude of $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}$. **[2]**

13. Find the distance between points $A(1, 2, 3)$ and $B(4, -1, 5)$. **[3]**

14. Position vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$, $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$. Find $|\mathbf{a} - \mathbf{b}|$. **[3]**

15. Show that $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$ are perpendicular. **[3]**

Section B – Hard (Q16–25)

16. Find the area of triangle ABC with position vectors $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$. **[7]**

17. Find the area of triangle with vertices $A(1, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, 3)$. **[6]**

18. Line $l : \mathbf{r} = \mathbf{i} + 2\mathbf{j} - \mathbf{k} + t(3\mathbf{i} - \mathbf{j} + 2\mathbf{k})$. Find the point on l closest to the origin. **[6]**

19. Determine whether $C(8, 5, -6)$ lies on the line through $A(2, -1, 4)$ and $B(5, 2, -1)$. **[5]**

20. Find the Cartesian equation of the line through $(1, 2, 3)$ parallel to $2\mathbf{i} - \mathbf{j} + \mathbf{k}$. **[4]**

21. Two lines: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}_1$ and $\mathbf{r} = \mathbf{b} + \mu\mathbf{d}_2$. Given they intersect and are perpendicular, find the intersection point (use given vectors). **[7]**

22. Find a vector perpendicular to both $\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. **[5]**

23. The points A, B, C, D have position vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}, \mathbf{i} + \mathbf{j} + \mathbf{k}$. Show that AB is perpendicular to CD. **[5]**

24. Find the angle between the lines $\mathbf{r} = (\mathbf{i} + \mathbf{j}) + \lambda(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = (2\mathbf{i} - \mathbf{j}) + \mu(2\mathbf{i} + \mathbf{j} - 2\mathbf{k})$. **[6]**

25. Position vector of A is $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$. Point B is such that $\overrightarrow{AB} = 3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$. Find the position vector of B and $|\overrightarrow{AB}|$. **[5]**

End of Worksheet • *Show all working clearly.*